

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

July 2022

Course Code: MAT102R01

Course: ENGINEERING MATHEMATICS - II

Question Paper No. :U0549

Duration: 3 hours

Max. Marks:100

PART-A

Answer all the questions

10 x 2 = 20 Marks

1. Find the envelope of the family of lines $y = mx + \sqrt{a^2 m^2 + b^2}$, m is parameter.
2. Determine whether (2,0) is minimum point of $u = x^3 + 3xy^2 - 3x^2 - 3y^2 + 7$.
3. Convert $\left(D^2 + \frac{2}{x}D\right)y = \frac{1}{x^4}$ into another differential equation with constant coefficient using suitable transformation.
4. Determine 'a' given that $(ay^2 - x^2)dx + 2xydy = 0$ is an exact equation.
5. If $\vec{A} = \nabla(xy + yz + zx)$, find $\nabla \times \vec{A}$.
6. Find the unit vector normal to the surface $x^2y + 2xz^2 = 8$ at the point (1,0,2).
7. Find the work done in moving a particle under the force field $(x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$ from (1,1,0) to (2,0,1).

8. State Green's theorem in plane.
9. If the temperature of the air is 40°C and the substance cool from 80°C to 60°C in 20 minutes. What will be the temperature of the substance after 40 minutes?
10. Show that the curves $y^2 = 4(x + 1)$ and $y^2 = 36(9 - x)$ cut orthogonally at the point (8,6).

PART-B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Find the radius of curvature of the curve $xy^2 = a(x^2 + y^2)$ at $(-2a, 2a)$.
- (b) Determine the radius of curvature of the curve $r^m = a^m \cos m\theta$.
(8+7)

(OR)

12. (a) Find the envelope of family of curves $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$, where a and b are connected by $a+b=c$.
- (b) A wire of length B is cut into two parts which are bent in the form of a square and circle respectively. Find the least value of the sum of the areas so formed.
(8+7)

13. (a) Solve $y \log y dx + (x - \log y) dy = 0$.
- (b) Solve $(D^3 + 3D^2 - 4D - 12)y = 12xe^{-2x}$
(8+7)

(OR)

14. (a) Solve $(4x^2 D^2 + 1)y = 19 \cos(\log x) + 22 \sin(\log x)$.
- (b) Applying method of variation of parameters, solve $(D^2 - 1)y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$.
(8+7)

15. (a) Find the directional derivative of $\phi = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of the normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$.

(b) Prove that $\nabla \cdot (r \nabla (\frac{1}{r^n})) = \frac{n(n-2)}{r^{n+1}}$. (8+7)

(OR)

16. (a) The temperature of the points in space is given by $\phi = x^2 + y^2 - z$. A mosquito located at the point $(1, 1, 2)$ desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move?

(b) Determine the value $\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$, where $\vec{A} = x^2 y \vec{i} + x^3 y^2 \vec{j} - 3x^2 z^2 \vec{k}$. (8+7)

17. (a) Applying Gauss divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ and S is the solid enclosed by $x+y+z=1$, $x=0$, $y=0$ and $z=0$.

(b) Applying Green's theorem, evaluate $\int_C \{(x-y)dx + 3xydy\}$, where C is the boundary of the region bounded by the parabolas $x^2 = 4y$ and $y^2 = 4x$. (8+7)

(OR)

18. Verify stoke's theorem for $\vec{F} = (x+y)\vec{i} + (y+z)\vec{j} - x\vec{k}$ and S is the surface of the plane $2x+y+z=2$ which is in the first octant.